

The Co_0 Leech-Space Sigma–MGF

Frame-shape proof, central parity, and a 24-dimensional stress test

Computational verification note

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Abstract

We prove the frame-shape class formula for the natural 24-dimensional complexified Leech-lattice representation of Co_0 and the vanishing of all odd total-degree coefficients. Since direct enumeration of Co_0 is not a reasonable notebook computation, we test the determinant mechanism at the correct dimension on the explicit signed-coordinate subgroup $\{\pm P_g : g \in M_{24}\}$. Direct invariant counts and exact class expansion agree through degree six. The distinction between this stress test and a full Co_0 class-table evaluation is kept explicit.

1 Frame-shape formula

Let $V = \Lambda_{\text{Leech}} \otimes_{\mathbb{Z}} \mathbb{C}$. If a conjugacy-class representative $g \in Co_0$ has frame shape $\pi_g = \prod_{r \geq 1} r^{k_r(g)}$, by definition

$$\det(xI - g) = \prod_{r \geq 1} (x^r - 1)^{k_r(g)}, \quad \sum_r r k_r(g) = 24.$$

Substitute $x = t^{-1}$ and multiply by t^{24} to obtain

$$\det(I - tg) = \prod_{r \geq 1} (1 - t^r)^{k_r(g)}.$$

The generalized Molien formula, grouped by conjugacy class, is therefore

$$\boxed{\mathcal{S}_{Co_0, V}(\mathbf{t}) = \sum_{C \subseteq Co_0} \frac{|C|}{|Co_0|} \prod_{i, r} (1 - t_i^r)^{-k_r(C)}}. \quad (1)$$

Negative frame exponents cause no difficulty: each class contribution is a rational function whose product is the characteristic determinant.

2 Central parity theorem

The central element $-I$ belongs to Co_0 . On $\text{Sym}^\tau V = \bigotimes_j \text{Sym}^{\tau_j} V$ it acts by $(-1)^{|\tau|}$. A fixed vector in odd total degree would satisfy $v = -v$, hence vanish. Thus

$$\boxed{[m_\tau] \mathcal{S}_{Co_0, V} = 0 \quad \text{whenever } |\tau| \text{ is odd.}} \quad (2)$$

In particular $[m_{(1)}] = 0$, unlike the M_{24} permutation representation, whose all-ones vector is fixed. The two 24-dimensional series therefore differ in their first positive degree.

This proof is representation theoretic and does not require any frame-shape table. As a data-integrity check on a full table, the class contributions in every odd degree must cancel exactly.

3 A tractable subgroup at the true dimension

In the standard coordinate construction of the Leech lattice, its monomial automorphism group contains M_{24} coordinate permutations and the central sign. We use

$$H = \{\pm P_g : g \in M_{24}\} \leq Co_0.$$

The notebook explicitly constructs three 24×24 integer orthogonal generators $P_a, P_b, -I$, using the ATLAS permutations a, b documented in the companion M_{24} note. This is large enough to exercise the actual ambient dimension while retaining an exact aggregated class computation.

If g has cycle counts $c_r(g)$, then

$$\det(I - tP_g) = \prod_r (1 - t^r)^{c_r(g)}, \quad \det(I + tP_g) = \prod_r (1 - (-t)^r)^{c_r(g)}.$$

Averaging the two signs projects onto even total degree. Hence

$$\mathcal{S}_H(t) = \frac{1}{2|M_{24}|} \sum_{g \in M_{24}} \left\{ \prod_r (1 - t^r)^{-c_r(g)} + \prod_r (1 - (-t)^r)^{-c_r(g)} \right\}. \quad (3)$$

4 Independent verification

For even k , $-I$ acts trivially on $\text{Sym}^k V$, so the H -fixed space has a basis indexed by M_{24} orbits of degree- k multisets. For odd k , (2) makes the dimension zero. This gives a direct generator-orbit algorithm independent of the class expansion (3). Exact results are:

k	direct invariant dimension	signed class formula	parity
1	0	0	odd
2	2	2	even
3	0	0	odd
4	5	5	even
5	0	0	odd
6	12	12	even

The degree-six value also exercises the first point at which the underlying M_{24} multiset orbits differ from those of S_{24} .

5 What this does and does not verify

The proof of (1) is exact for full Co_0 , and the proof of (2) establishes every odd-degree zero for full Co_0 . The matrix sweep validates determinant expansion, sign cancellation, and direct invariant counting in dimension 24 on a genuine subgroup. It does not claim that the even coefficients 2, 5, 12 are full- Co_0 coefficients; fixed spaces can shrink when the acting group grows. A full numerical evaluation of (1) requires the complete Co_0 frame-shape and class-size table, which is deliberately not reconstructed from partial data.

Run

```
.venv/bin/python for_this_guy/sporadic_matrix_groups/
co0_leech/verification.py
.venv/bin/marimo edit for_this_guy/sporadic_matrix_groups/
co0_leech/notebook.py
```

after joining wrapped lines. For background on the Leech representation and Conway fixed lattices, see <https://arxiv.org/abs/1505.06420>.